DATA ANALYTICS (DA)

EXCEL ASSIGNMENT 2

Dataset: Product Dataset

Software: Microsoft Excel

Objective

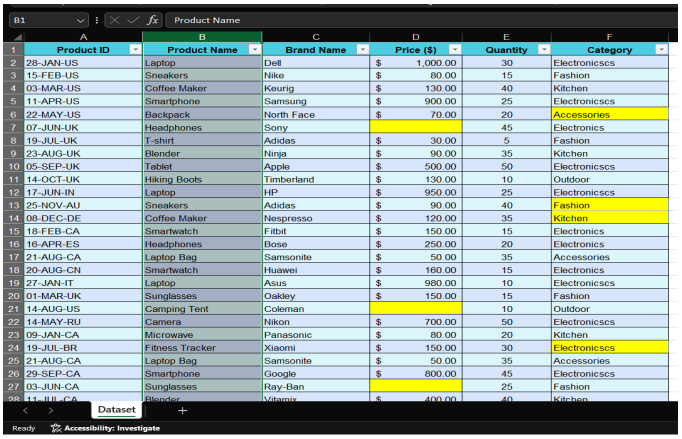
To identify missing values in the Price and Category columns and apply appropriate data-cleaning methods so that the dataset is more complete and reliable for analysis.

1. Handling Missing Values:

Product ID	Product Name	Brand Name	Price ($)	Quantity	Category
28-JAN-US	laptop	Dell	$ 1,000.00	30	Electronics
15-FEB-US	Sneakers	Nike	$ 80.00	15	Fashion
03-MAR-US	Coffee Maker	Keurig	$ 130.00	40	Kitchen
11-APR-US	smartphone	Samsung	$ 900.00	25	Electronics
22-MAY-US	Backpack	North Face	$ 70.00	20	Accessories
07-JUN-UK	Headphones	Sony		45	Electroni
19-JUL-UK	T-shirt	Adidas	$ 30.00	5	Fashion
23-AUG-UK	Blender	Ninja	$ 90.00	35	Kitchen
05-SEP-UK	Tablet	Apple	$ 500.00	50	Electronics
14-OCT-UK	Hiking Boots	Timberland	$ 130.00	10	Outdoor
17-JUN-IN	Laptop	HP	$ 950.00	25	Electronics
25-NOV-AU	Sneakers	Adidas	$ 90.00	40	Fashion
08-DEC-DE	Coffee Maker	Nespresso	$ 120.00	35	Kitchen
18-FEB-CA	Smartwatch	Fitbit	$ 150.00	15	Electroni
16-APR-ES	headphones	Bose	$ 250.00	20	Electroni
21-AUG-CA	Laptop Bag	Samsonite	$ 50.00	35	Accessories
20-AUG-CN	Smartwatch	Huawei	$ 160.00	15	Electroni
27-JAN-IT	laptop	Asus	$ 980.00	10	Electronics
01-MAR-UK	Sunglasses	Oakley	$ 150.00	15	Fashion
14-AUG-US	Camping Tent	Coleman		10	Outdoor
14-MAY-RU	Camera	Nikon	$ 700.00	50	Electronics
09-JAN-CA	Microwave	Panasonic	$ 80.00	20	Kitchen
19-JUL-BR	Fitness Tracker	Xiaomi	$ 150.00	30	Electronics
21-AUG-CA	Laptop Bag	Samsonite	$ 50.00	35	Accessories
29-SEP-CA	Smartphone	Google	$ 800.00	45	Electronics
03-JUN-CA	Sunglasses	Ray-Ban		25	Fashion

2. Correcting Inconsistent Data:

~ ***Product Name Standardization***



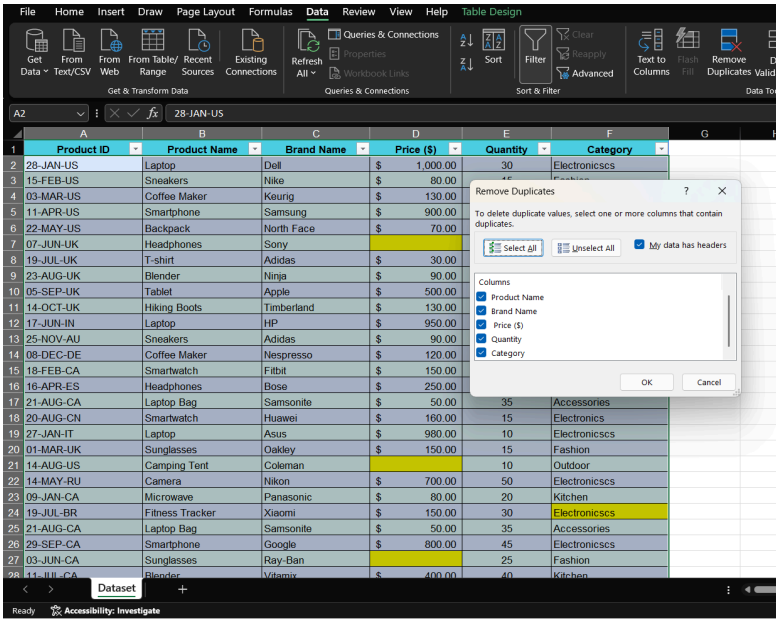
Product ID	Product Name	Brand Name	Price ($)	Quantity	Category
28-JAN-US	Laptop	Dell	$ 1,000.00	30	Electronicscs
15-FEB-US	Sneakers	Nike	$ 80.00	15	Fashion
03-MAR-US	Coffee Maker	Keurig	$ 130.00	40	Kitchen
11-APR-US	Smartphone	Samsung	$ 900.00	25	Electronicscs
22-MAY-US	Backpack	North Face	$ 70.00	20	Accessories
07-JUN-UK	Headphones	Sony		45	Electronics
19-JUL-UK	T-shirt	Adidas	$ 30.00	5	Fashion
23-AUG-UK	Blender	Ninja	$ 90.00	35	Kitchen
05-SEP-UK	Tablet	Apple	$ 500.00	50	Electronicscs
14-OCT-UK	Hiking Boots	Timberland	$ 130.00	10	Outdoor
17-JUN-IN	Laptop	HP	$ 950.00	25	Electronicscs
25-NOV-AU	Sneakers	Adidas	$ 90.00	40	Fashion
08-DEC-DE	Coffee Maker	Nespresso	$ 120.00	35	Kitchen
18-FEB-CA	Smartwatch	Fitbit	$ 150.00	15	Electronics
16-APR-ES	Headphones	Bose	$ 250.00	20	Electronics
21-AUG-CA	Laptop Bag	Samsonite	$ 50.00	35	Accessories
20-AUG-CN	Smartwatch	Huawei	$ 160.00	15	Electronics
27-JAN-IT	Laptop	Asus	$ 980.00	10	Electronicscs
01-MAR-UK	Sunglasses	Oakley	$ 150.00	15	Fashion
14-AUG-US	Camping Tent	Coleman		10	Outdoor
14-MAY-RU	Camera	Nikon	$ 700.00	50	Electronicscs
09-JAN-CA	Microwave	Panasonic	$ 80.00	20	Kitchen
19-JUL-BR	Fitness Tracker	Xiaomi	$ 150.00	30	Electronicscs
21-AUG-CA	Laptop Bag	Samsonite	$ 50.00	35	Accessories
29-SEP-CA	Smartphone	Google	$ 800.00	45	Electronicscs
03-JUN-CA	Sunglasses	Ray-Ban		25	Fashion

~ ***Category Correction***



Product ID	Product Name	Brand Name	Price ($)	Quantity	Category
28-JAN-US	Laptop	Dell	$ 1,000.00	30	Electronicscs
15-FEB-US	Sneakers	Nike	$ 80.00	15	Fashion
03-MAR-US	Coffee Maker	Keurig	$ 130.00	40	Kitchen
11-APR-US	Smartphone	Samsung	$ 900.00	25	Electronicscs
22-MAY-US	Backpack	North Face	$ 70.00	20	Accessories
07-JUN-UK	Headphones	Sony		45	Electronics
19-JUL-UK	T-shirt	Adidas	$ 30.00	5	Fashion
23-AUG-UK	Blender	Ninja	$ 90.00	35	Kitchen
05-SEP-UK	Tablet	Apple	$ 500.00	50	Electronicscs
14-OCT-UK	Hiking Boots	Timberland	$ 130.00	10	Outdoor
17-JUN-IN	Laptop	HP	$ 950.00	25	Electronicscs
25-NOV-AU	Sneakers	Adidas	$ 90.00	40	Fashion
08-DEC-DE	Coffee Maker	Nespresso	$ 120.00	35	Kitchen
18-FEB-CA	Smartwatch	Fitbit	$ 150.00	15	Electronics
16-APR-ES	Headphones	Bose	$ 250.00	20	Electronics
21-AUG-CA	Laptop Bag	Samsonite	$ 50.00	35	Accessories
20-AUG-CN	Smartwatch	Huawei	$ 160.00	15	Electronics
27-JAN-IT	Laptop	Asus	$ 980.00	10	Electronicscs
01-MAR-UK	Sunglasses	Oakley	$ 150.00	15	Fashion
14-AUG-US	Camping Tent	Coleman		10	Outdoor
14-MAY-RU	Camera	Nikon	$ 700.00	50	Electronicscs
09-JAN-CA	Microwave	Panasonic	$ 80.00	20	Kitchen
19-JUL-BR	Fitness Tracker	Xiaomi	$ 150.00	30	Electronicscs
21-AUG-CA	Laptop Bag	Samsonite	$ 50.00	35	Accessories
29-SEP-CA	Smartphone	Google	$ 800.00	45	Electronicscs
03-JUN-CA	Sunglasses	Ray-Ban		25	Fashion

3. Removing Duplicates:

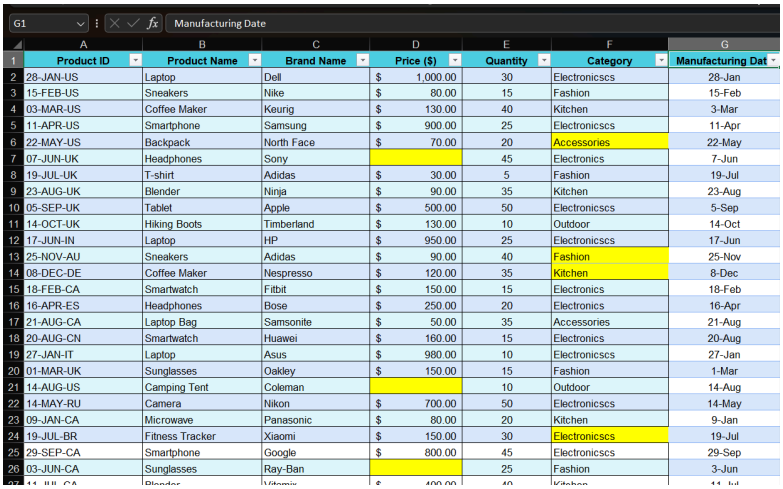


Result: *Three duplicate rows were identified and removed. The dataset now contains 31 unique records.*

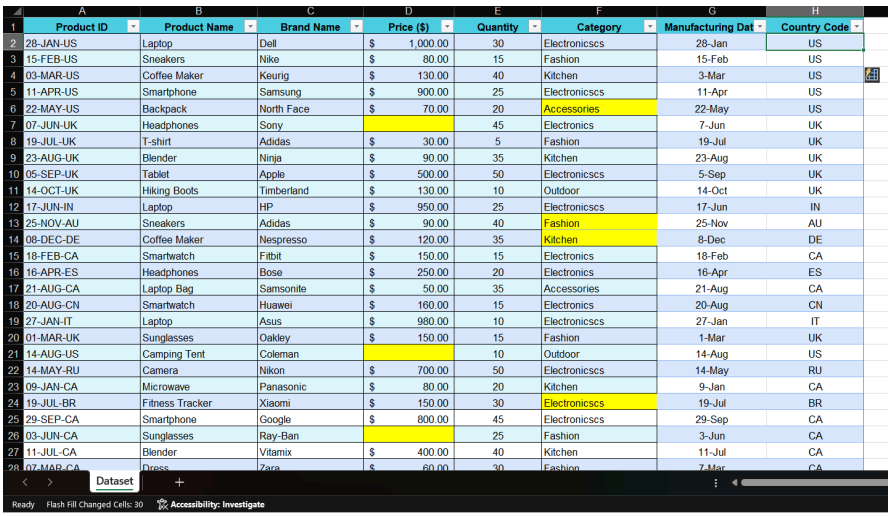
4 .Splitting and Merging Data:

Used Flash Fill to separate the Manufacturing Date and Country Code from the Product ID column and merge Brand Name and Product Name into a separate column named Product Brand.

~ ***Split Product ID***



Product ID	Product Name	Brand Name	Price ($)	Quantity	Category	Manufacturing Dat
28-JAN-US	Laptop	Dell	$ 1,000.00	30	Electronicscs	28-Jan
15-FEB-US	Sneakers	Nike	$ 80.00	15	Fashion	15-Feb
03-MAR-US	Coffee Maker	Keurig	$ 130.00	40	Kitchen	3-Mar
11-APR-US	Smartphone	Samsung	$ 900.00	25	Electronicscs	11-Apr
22-MAY-US	Backpack	North Face	$ 70.00	20	Accessories	22-May
07-JUN-UK	Headphones	Sony		45	Electronics	7-Jun
19-JUL-UK	T-shirt	Adidas	$ 30.00	5	Fashion	19-Jul
23-AUG-UK	Blender	Ninja	$ 90.00	35	Kitchen	23-Aug
05-SEP-UK	Tablet	Apple	$ 500.00	50	Electronicscs	5-Sep
14-OCT-UK	Hiking Boots	Timberland	$ 130.00	10	Outdoor	14-Oct
17-JUN-IN	Laptop	HP	$ 950.00	25	Electronicscs	17-Jun
25-NOV-AU	Sneakers	Adidas	$ 90.00	40	Fashion	25-Nov
08-DEC-DE	Coffee Maker	Nespresso	$ 120.00	35	Kitchen	8-Dec
18-FEB-CA	Smartwatch	Fitbit	$ 150.00	15	Electronics	18-Feb
16-APR-ES	Headphones	Bose	$ 250.00	20	Electronics	16-Apr
21-AUG-CA	Laptop Bag	Samsonite	$ 50.00	35	Accessories	21-Aug
20-AUG-CN	Smartwatch	Huawei	$ 160.00	15	Electronics	20-Aug
27-JAN-IT	Laptop	Asus	$ 980.00	10	Electronicscs	27-Jan
01-MAR-UK	Sunglasses	Oakley	$ 150.00	15	Fashion	1-Mar
14-AUG-US	Camping Tent	Coleman		10	Outdoor	14-Aug
14-MAY-RU	Camera	Nikon	$ 700.00	50	Electronicscs	14-May
09-JAN-CA	Microwave	Panasonic	$ 80.00	20	Kitchen	9-Jan
19-JUL-BR	Fitness Tracker	Xiaomi	$ 150.00	30	Electronicscs	19-Jul
29-SEP-CA	Smartphone	Google	$ 800.00	45	Electronicscs	29-Sep
03-JUN-CA	Sunglasses	Ray-Ban		25	Fashion	3-Jun

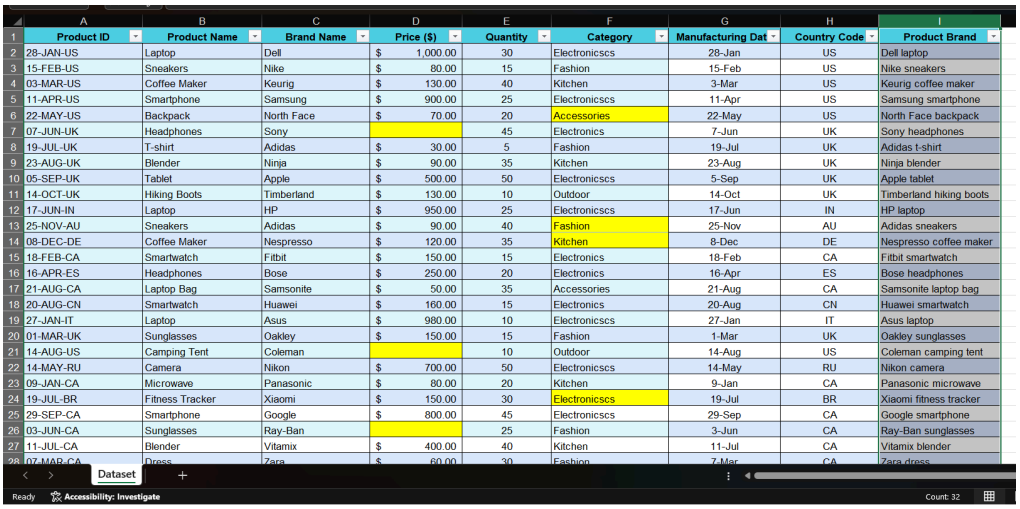


	A	B	C	D	E	F	G	H
1	Product ID	Product Name	Brand Name	Price ($)	Quantity	Category	Manufacturing Dat	Country Code
2	28-JAN-US	Laptop	Dell	$ 1,000.00	30	Electronicscs	28-Jan	US
3	15-FEB-US	Sneakers	Nike	$ 80.00	15	Fashion	15-Feb	US
4	03-MAR-US	Coffee Maker	Keurig	$ 130.00	40	Kitchen	3-Mar	US
5	11-APR-US	Smartphone	Samsung	$ 900.00	25	Electronicscs	11-Apr	US
6	22-MAY-US	Backpack	North Face	$ 70.00	20	Accessories	22-May	US
7	07-JUN-UK	Headphones	Sony		45	Electronics	7-Jun	UK
8	19-JUL-UK	T-shirt	Adidas	$ 30.00	5	Fashion	19-Jul	UK
9	23-AUG-UK	Blender	Ninja	$ 90.00	35	Kitchen	23-Aug	UK
10	05-SEP-UK	Tablet	Apple	$ 500.00	50	Electronicscs	5-Sep	UK
11	14-OCT-UK	Hiking Boots	Timberland	$ 130.00	10	Outdoor	14-Oct	UK
12	17-JUN-IN	Laptop	HP	$ 950.00	25	Electronicscs	17-Jun	IN
13	25-NOV-AU	Sneakers	Adidas	$ 90.00	40	Fashion	25-Nov	AU
14	08-DEC-DE	Coffee Maker	Nespresso	$ 120.00	35	Kitchen	8-Dec	DE
15	18-FEB-CA	Smartwatch	Fitbit	$ 150.00	15	Electronics	18-Feb	CA
16	16-APR-ES	Headphones	Bose	$ 250.00	20	Electronics	16-Apr	ES
17	21-AUG-CA	Laptop Bag	Samsonite	$ 50.00	35	Accessories	21-Aug	CA
18	20-AUG-CN	Smartwatch	Huawei	$ 160.00	15	Electronics	20-Aug	CN
19	27-JAN-IT	Laptop	Asus	$ 980.00	10	Electronicscs	27-Jan	IT
20	01-MAR-UK	Sunglasses	Oakley	$ 150.00	15	Fashion	1-Mar	UK
21	14-AUG-US	Camping Tent	Coleman		10	Outdoor	14-Aug	US
22	14-MAY-RU	Camera	Nikon	$ 700.00	50	Electronicscs	14-May	RU
23	09-JAN-CA	Microwave	Panasonic	$ 80.00	20	Kitchen	9-Jan	CA
24	19-JUL-BR	Fitness Tracker	Xiaomi	$ 150.00	30	Electronicscs	19-Jul	BR
25	29-SEP-CA	Smartphone	Google	$ 800.00	45	Electronicscs	29-Sep	CA
26	03-JUN-CA	Sunglasses	Ray-Ban		25	Fashion	3-Jun	CA
27	11-JUL-CA	Blender	Vitamix	$ 400.00	40	Kitchen	11-Jul	CA
28	07-MAR-CA	Dress	Zara	$ 60.00	30	Fashion	7-Mar	CA

Dataset

Ready Flash Fill Changed Cells: 30 Accessibility: Investigate

~ ***Merge Brand Name and Product Name***



	A	B	C	D	E	F	G	H	I
1	Product ID	Product Name	Brand Name	Price ($)	Quantity	Category	Manufacturing Dat	Country Code	Product Brand
2	28-JAN-US	Laptop	Dell	$ 1,000.00	30	Electronicscs	28-Jan	US	Dell laptop
3	15-FEB-US	Sneakers	Nike	$ 80.00	15	Fashion	15-Feb	US	Nike sneakers
4	03-MAR-US	Coffee Maker	Keurig	$ 130.00	40	Kitchen	3-Mar	US	Keurig coffee maker
5	11-APR-US	Smartphone	Samsung	$ 900.00	25	Electronicscs	11-Apr	US	Samsung smartphone
6	22-MAY-US	Backpack	North Face	$ 70.00	20	Accessories	22-May	US	North Face backpack
7	07-JUN-UK	Headphones	Sony		45	Electronics	7-Jun	UK	Sony headphones
8	19-JUL-UK	T-shirt	Adidas	$ 30.00	5	Fashion	19-Jul	UK	Adidas t-shirt
9	23-AUG-UK	Blender	Ninja	$ 90.00	35	Kitchen	23-Aug	UK	Ninja blender
10	05-SEP-UK	Tablet	Apple	$ 500.00	50	Electronicscs	5-Sep	UK	Apple tablet
11	14-OCT-UK	Hiking Boots	Timberland	$ 130.00	10	Outdoor	14-Oct	UK	Timberland hiking boots
12	17-JUN-IN	Laptop	HP	$ 950.00	25	Electronicscs	17-Jun	IN	HP laptop
13	25-NOV-AU	Sneakers	Adidas	$ 90.00	40	Fashion	25-Nov	AU	Adidas sneakers
14	08-DEC-DE	Coffee Maker	Nespresso	$ 120.00	35	Kitchen	8-Dec	DE	Nespresso coffee maker
15	18-FEB-CA	Smartwatch	Fitbit	$ 150.00	15	Electronics	18-Feb	CA	Fitbit smartwatch
16	16-APR-ES	Headphones	Bose	$ 250.00	20	Electronics	16-Apr	ES	Bose headphones
17	21-AUG-CA	Laptop Bag	Samsonite	$ 50.00	35	Accessories	21-Aug	CA	Samsonite laptop bag
18	20-AUG-CN	Smartwatch	Huawei	$ 160.00	15	Electronics	20-Aug	CN	Huawei smartwatch
19	27-JAN-IT	Laptop	Asus	$ 980.00	10	Electronicscs	27-Jan	IT	Asus laptop
20	01-MAR-UK	Sunglasses	Oakley	$ 150.00	15	Fashion	1-Mar	UK	Oakley sunglasses
21	14-AUG-US	Camping Tent	Coleman		10	Outdoor	14-Aug	US	Coleman camping tent
22	14-MAY-RU	Camera	Nikon	$ 700.00	50	Electronicscs	14-May	RU	Nikon camera
23	09-JAN-CA	Microwave	Panasonic	$ 80.00	20	Kitchen	9-Jan	CA	Panasonic microwave
24	19-JUL-BR	Fitness Tracker	Xiaomi	$ 150.00	30	Electronicscs	19-Jul	BR	Xiaomi fitness tracker
25	29-SEP-CA	Smartphone	Google	$ 800.00	45	Electronicscs	29-Sep	CA	Google smartphone
26	03-JUN-CA	Sunglasses	Ray-Ban		25	Fashion	3-Jun	CA	Ray-Ban sunglasses
27	11-JUL-CA	Blender	Vitamix	$ 400.00	40	Kitchen	11-Jul	CA	Vitamix blender
28	07-MAR-CA	Dress	Zara	$ 60.00	30	Fashion	7-Mar	CA	Zara dress

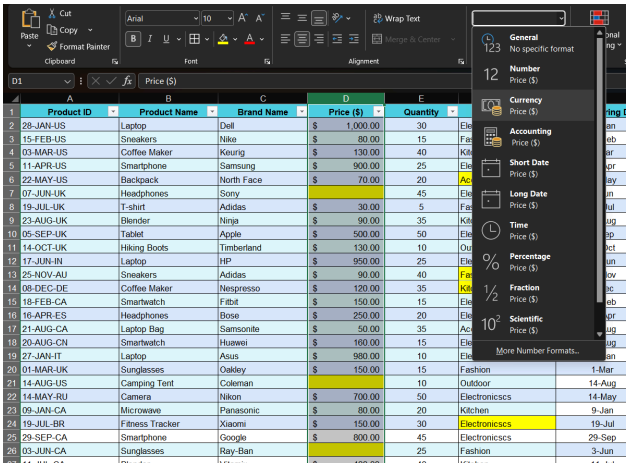
Dataset

Ready Accessibility: Investigate Count: 32

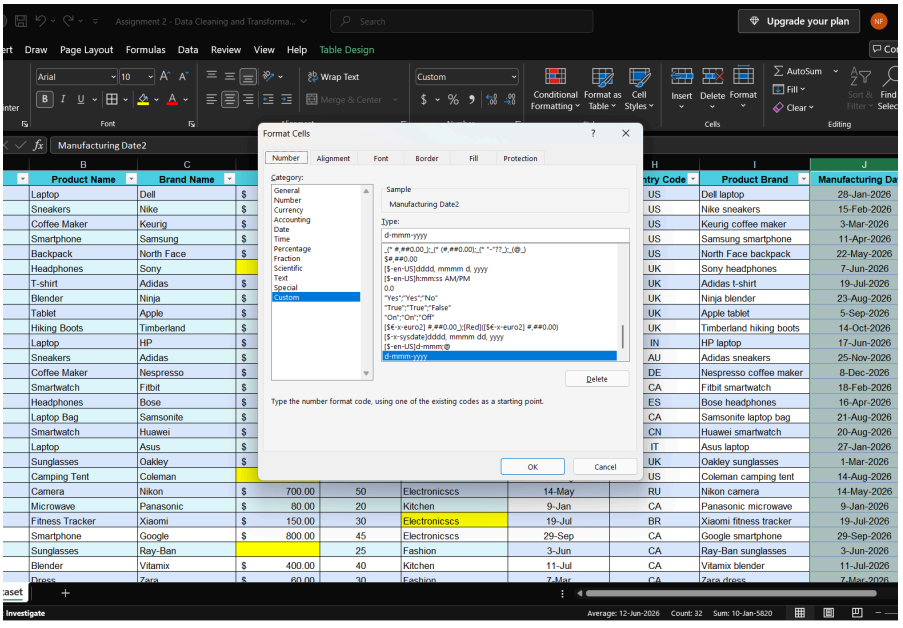
Result: *The Product ID was separated into two columns, and the Brand Name and Product Name were combined into one column using Flash Fill.*

5. Number Formatting

~ ***Format Price as Currency***



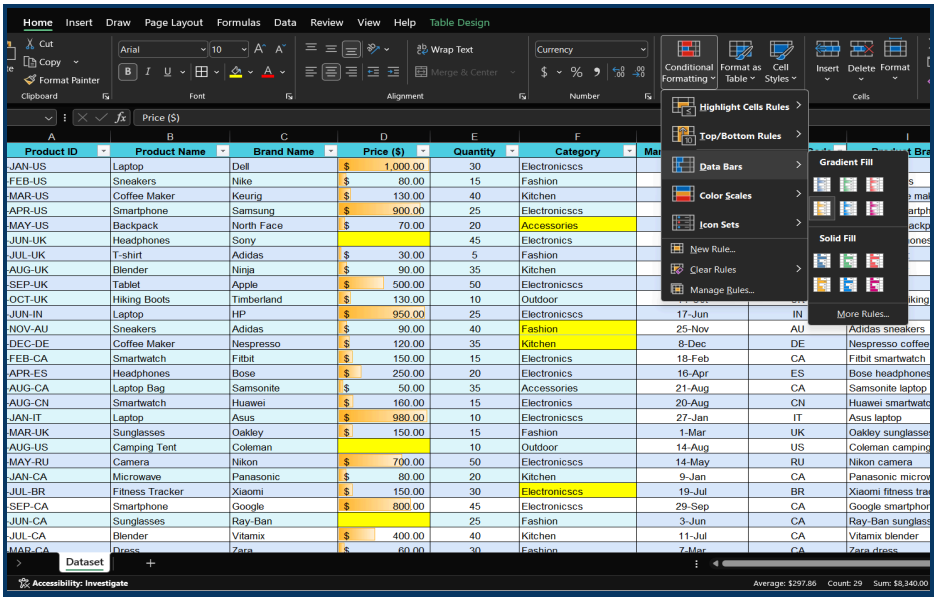
~ ***Format Manufacturing Date***



***Result:** The Price column is displayed in currency format, and the Manufacturing Date column is displayed in the required date format.*

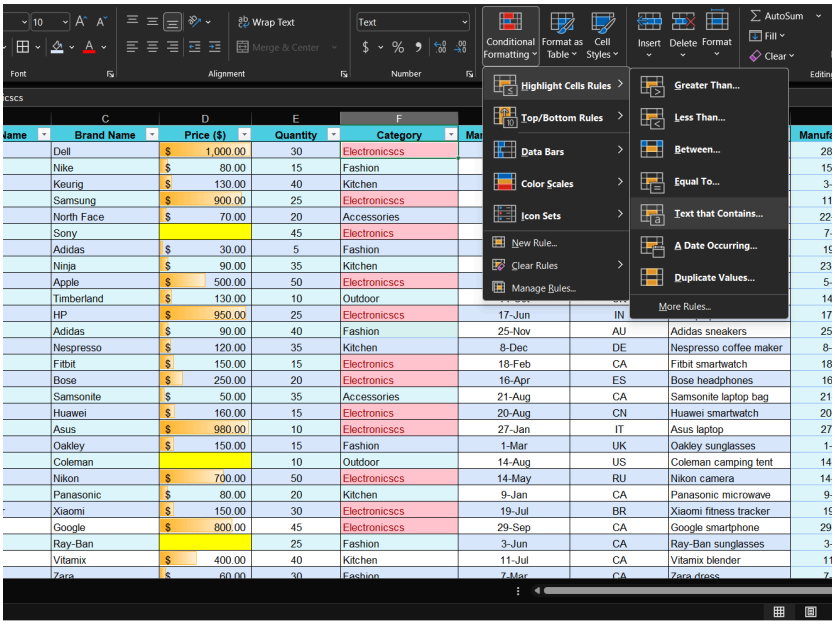
6. Conditional Formatting

~ Apply Data Bars to the Price column



Result: *Higher prices will have longer data bars, and lower prices will have shorter data bars.*

~ Highlight Electronics in the Category column



Result: *Data Bars were applied to the Price column to compare prices visually. All Electronics category cells were highlighted using a custom conditional formatting rule.*